Geometry
Unit 9
Calculator

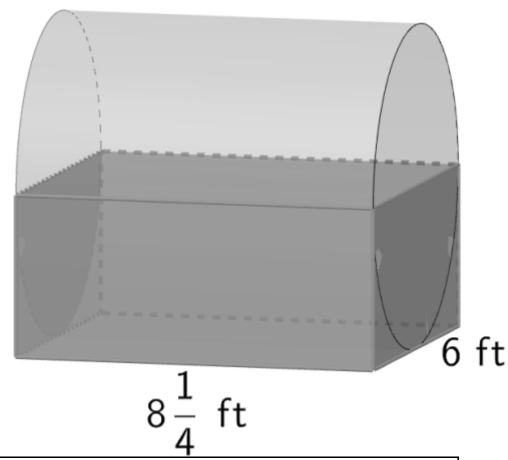
Name____________________
Date_________________
Period__________

Prism/Cylinder	Cone	Regular Pyramid	Spheres
$SA = 2B + Ph$ $V = Bh$	$SA = B + \pi r h_s$ $V = \frac{1}{3}Bh$	$SA = B + \frac{1}{2}Ph_s$ $V = \frac{1}{3}Bh$	$SA = 4\pi r^2$ $V = \frac{4}{3}\pi r^3$

1. A carpenter is designing a child's toy chest. He creates a drawing using a rectangular prism and one half of a cylinder. His drawing is shown with measurements in feet (ft).

Determine the surface area of the toy chest.

Show your work and include units in the answer. Round your answer to the nearest thousandth.



work	
Surface Area	241.029 ft^2

2. A stack of 6 hexagonal blocks is shown, where the base area of each block is 32.56 square inches (in.^{2}) and has a thickness of 0.12 inches (in.).

A stack of 6 square blocks is shown, where the side length of each square block is 5.75 inches (in.) and has a thickness of 0.12 in.

Complete the statements. Show your work to justify your selections.

The volume of the stack of hexagonal blocks is

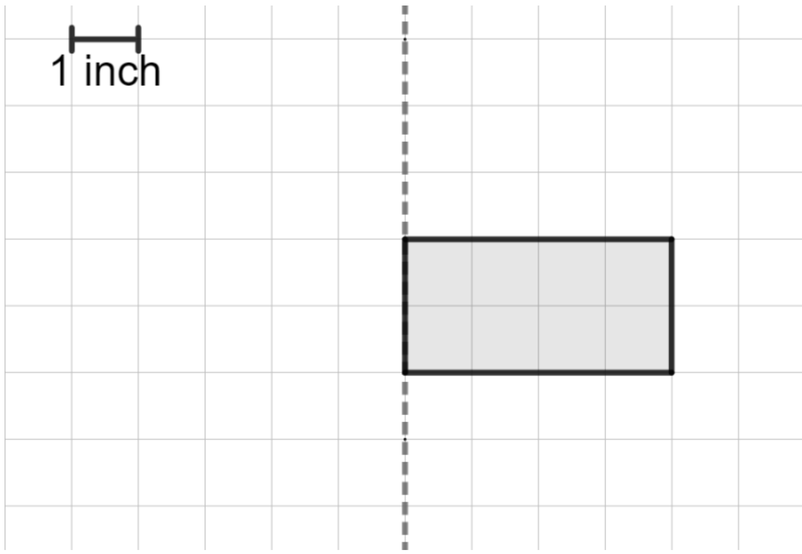
- ○ equal to
- ○ less than
- ○ greater than

the volume of the stack of square blocks. This is because

- ○ both blocks have the same thickness.
- ○ both stacks have the same number of blocks.
- ○ the base area of the hexagon block is greater than the base area of the square block.
- ○ the base area of the square block is greater than the base area of the hexagon block.

work

3. A rectangle with dimensions of 2 inches (in.) by 4 in. is shown.



Complete the statement.

When the rectangle is rotated about the dotted line, the resulting figure is a

- cone
- cylinder
- square pyramid
- rectangular prism

with a height of

- 2 in.
- 4 in.
- 8 in.

and a

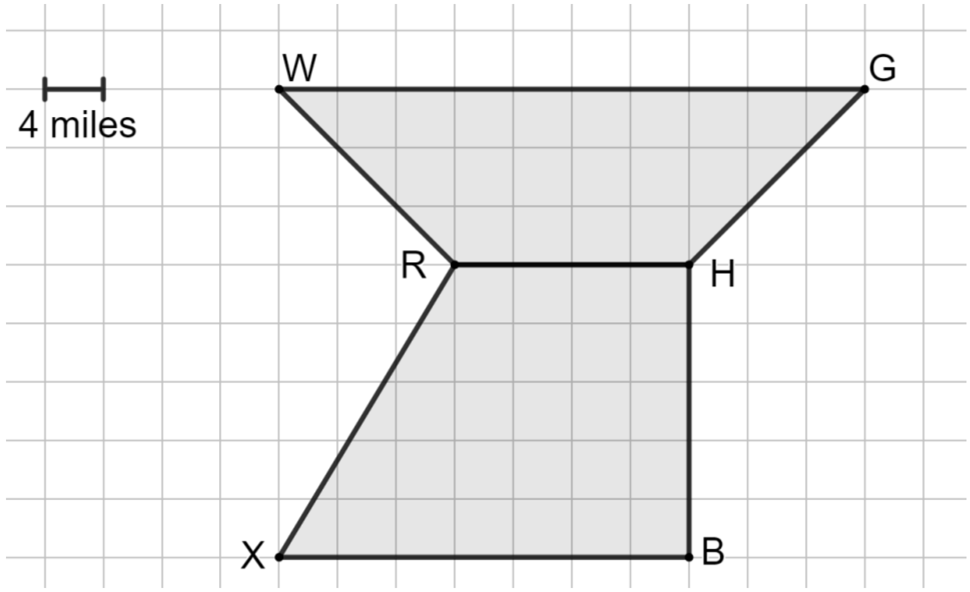
- radius of 2 in.
- radius of 4 in.
- radius of 8 in.
- length of 2 in.
- length of 4 in.
- length of 8 in.

4. Determine the lateral area of a triangular prism with a height of 20 centimeters (cm) and whose base is a right triangle with side lengths of 8 cm, 15 cm, and 17 cm.

Show your work and include units in the answer.

work	
Lateral Area	800 cm^2

5. Each year Sean and his family fish for lobsters during Florida's mini-season, a two-day event typically in July. Sean uses the diagram shown to note the areas he likes to fish.



a. Find the total area, in square miles, of the areas that Sean likes to fish. Show your work and include units in the answer.

work	
Area	776 mi^2

b. Complete the statement.

If Sean changed the scale of his diagram from 4 miles to 1 mile the sides of the quadrilaterals will

- ○ decrease by a factor of $\frac{1}{4}$.
- ○ decrease by a factor of 4.
- ○ increase by a factor of $\frac{1}{4}$.
- ○ increase by a factor of 4.

c. Prior to going, Sean read that the lobster population in the area labeled $WGHR$ is estimated to be $3{,}091$ lobsters.

Determine the density of the lobster population in the area labeled $WGHR$. Show your work and include units in the answer. Round to the nearest tenth.

work	
Density	9.2 lobsters/mi^2

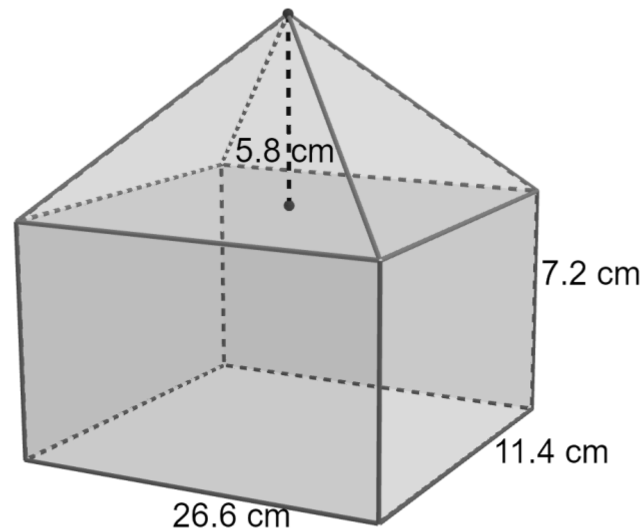
6. Find the volume of a sphere whose radius is 16.5 centimeters (cm).

Show your work and include units in the answer. Leave your answer in π form.

work	
Volume	5989.5π cm^3

7. A composite figure is shown with measurements in centimeter (cm). Find the volume of the composite figure.

Show your work and include units in the answer. Round to the nearest thousandth.



work	
Volume	2769.592 cm^3

8. Select all of the descriptions that would result in rectangular cross sections when the given plane slices through the given figure.

- ☐ A plane parallel to the base of a cone
- ☐ A plane perpendicular to the base of a cylinder
- ☐ A plane parallel to the base of a triangular prism
- ☐ A plane parallel to the base of a rectangular prism
- ☐ A plane perpendicular to the base of a triangular prism
- ☐ A plane perpendicular to the base of a rectangular pyramid